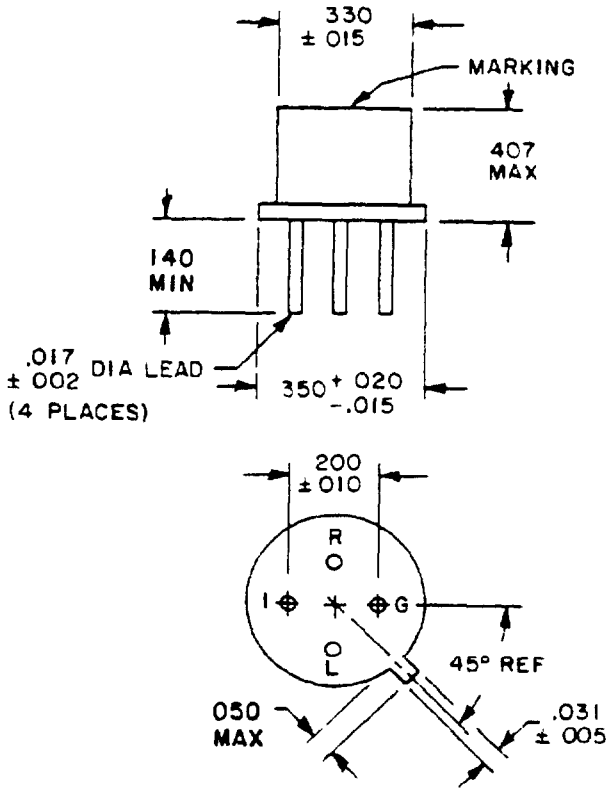
MILITARY SPECIFICATION SHEET

MIXER STAGES, RADIO FREQUENCY, DOUBLE BALANCED, TO-5 CONFIGURATION

This specification is approved for use by all Departments and Agencies of the Department of Defense

The complete requirements for procuring mixers described herein shall consist of this document and the latest issue of Specification MIL-M-28837.



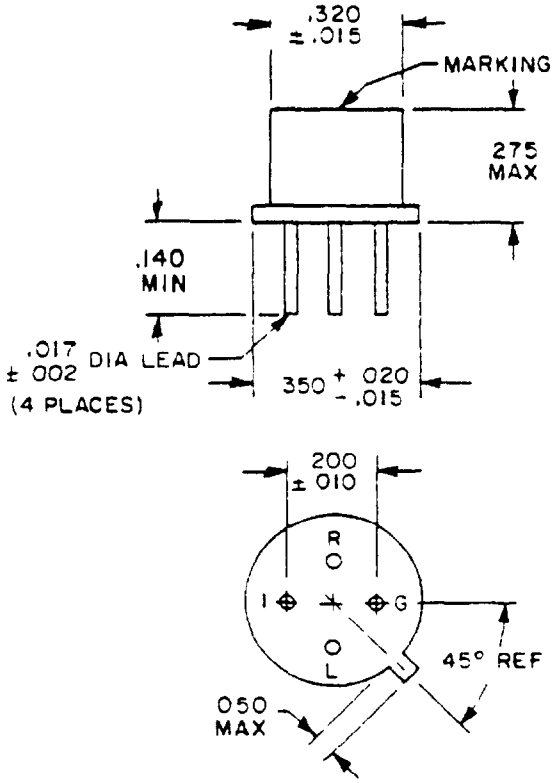
INCHES	MM
.002	.05
.005	.13
.010	.25
015	.38
.017	43
.020	51
025	64
031	79
050	1 27
140	3 56
.200	5 08
.330	8 38
.350	8 89
407	10 34

NOTES
1. Dimensions are in inches
2 Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 1. Outline drawing for mixers P/N M28837/7-01N -02N and -02S.

FSC 5985

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INCHES	MM
002	.05
010	25
.015	.38
017	43
020	51
050	1.27
140	3 56
.200	5.08
275	6.98
320	8 13
350	8 89

NOTES
1 Dimensions are in inches
2 Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm

FIGURE 2. Outline drawing for mixer P/N M28837/7-03N

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REQUIREMENTS:

Design and construction:
Dimensions and configuration. See figures 1 and 2.

Electrical characteristics:
Operating frequency range: See table I
LO drive power: See table I.
Conversion loss (max): See table I.
Noise figure (SSB): See table I.
Isolation (minimum): See table I.
Maximum input power: See table I
Conversion compression. See table I.
Desensitization. See table I.

Physical and Environmental characteristics:
Weight: 2 grams maximum.
Temperature. -01N, 03N = operating: -54° to +100°C.
nonoperating -65° to +100°C.
-02N, 02S = operating -65° to +100°C.
nonoperating: -65° to +100°C.
Shock. Test condition C.
Life: -01 = 100,000 hours minimum.
Seal. Hermetic.

Part number: M28837/7 - (dash number from table I).

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TABLE I Electrical characteristics

Dash no	Operating frequency range	LO input drive power in dBm 1/	Maximum conversion loss (SSB) dB MHz	Noise figure SSB dB MHz	ISOLATION Minimum LO-RF dB	LO-IF dB	RF-IF dB	Frequency MHz
01N	RF 10-500 MHz LO 10-500 MHz IF DC-500 MHz	nominal +7 TV +7	7 at f_L and f_R 10-200 f_I DC - 200 8 at f_L and f_R 200-350 f_I DC - 350 9 at f_L and f_R 350-500 f_I DC - 500	7 at f_L and f_R 10-200 f_I 0 4-200 8 at f_L and f_R 200-350 f_I 0 4-350 9 at f_L and f_R 350-500 f_I 0 4 - 500	40 35 30 25	35 30 25 15	25 20 15 10	10-50 50-100 100-200 200-500
02N 2/ 02S	RF 350-500 MHz LO 350-500 MHz IF DC-500 MHz	nominal +7 TV +7	9 at f_L and f_R 350-500 f_I DC - 500	9 at f_L and f_R 350-500 f_I 0 4 - 500	25	15	10	350-500
03N	RF 0 4-500 MHz LO 0 4-500 MHz IF DC-500 MHz	minimum +4 TV +7 maximum +13	6 5 at f_L and f_R 1-200 f_I DC - 200 7 5 at f_L and f_R 0 4-500 f_I DC - 500	6 5 at f_I and f_R 2-200 f_I 2 - 200 7 5 at f_L and f_R 2-500 f_I 2 - 500	45 30	30 20	N/A N/A	0 4-60 60 500

1/ TV is the test value of the LO power for electrical characteristics unless otherwise specified
2/ Use for reprocurement purposes only For new design see MIL-STD-1640
3/ With two in phase signals applied to the LO and RF ports

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NOTE: Mixers covered by this specification sheet are substituted for the following manufacturer's part number. This information in no way implies that the manufacturer's part is suitable as a substitute for the military part number.

Dash number	FSN	Manufacturer's code	Commercial type.
01N	---	05869	723880-4
01N	---	14482	M6T-1
02N	---	---	---
02S	---	97942	583R506H01
03N	---	14482	M6V

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - EC, OS
Air Force - 11, 17, 99
DLA - ES

User activities
Army - AR
Navy - AS, CG, MC
Air Force - 14, 19

Preparing activity
Navy - EC

Agent
DLA - ES

(Project 5985-0886-3)

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No 22 R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-M-28837/7

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?
☐ YES ☐ NO *(If "Yes" in what way?)*

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SUBMITTED BY *(Printed or typed name and address - Optional)*

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